

Dominic Bunnett

dombunnett.github.io

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Research interests

Moduli theory and geometric invariant theory, at the interface of algebraic and discrete geometry, with an emphasis on computational and algorithmic problems.

Positions

- **Postdoc**, research group Algorithmic Algebra (Peter Bürgisser), Technische Universität Berlin, October 2021 – present
- **Associate Lecturer**, The Open University, UK, September 2024 – present
- **Lecturer**, Makerere University, Kampala, April 2022 – January 2023
- **Postdoc**, Technische Universität Berlin, August 2019 – May 2021, within the Einstein Visiting Fellowship of Bernd Sturmfels (mentors: Michael Joswig, Bernd Sturmfels)

Education

- **PhD in Mathematics**, Freie Universität Berlin, 2015 – 2019. Thesis: *Moduli of hypersurfaces in toric orbifolds*. Advisor: Victoria Hoskins
- **BSc and MMath**, University of Warwick, 2011 – 2015, 1st class. Advisor: Miles Reid

Publications

- On the moduli of hypersurfaces in toric orbifolds. *Proc. Edinburgh Math. Soc.*, 67(2):577–616, 2024. doi:10.1017/S0013091524000166
- Generalised cone complexes and tropical moduli in polymake. *With M. Joswig and J. Pfeifle*. In *Proc. ISSAC '23*, 100–106. ACM, 2023. doi:10.1145/3597066.3597093
- Determinantal representations of the cubic discriminant. *With H. Keneshlou*. *Le Matematiche*, 75(2):489–505, 2020.

Preprints

- Embeddings of weighted projective spaces. *With P. Adeyemo and F. Levicán-Santibáñez*. arXiv:2510.05076
 - Moduli of Bridgeland semistable holomorphic triples. *With A. Rincón-Hidalgo*. arXiv:2102.04995
- Four further papers in preparation, on Bott vanishing and finiteness of automorphisms (with C. Namanya), K-moduli of exceptional del Pezzo hypersurfaces, moduli stacks of hypersurfaces, and Hilbert–Mumford criteria.

Awards and competitive research programmes

- **Research in Teams**, Erwin Schrödinger Institute, Vienna, 2025. Project *Projective geometry, combinatorics and representation theory from polytopes*, with P. Adeyemo, L. Campo and B. Szendrői
- **Research in Pairs**, Mathematisches Forschungsinstitut Oberwolfach, 2021, with A. Rincón-Hidalgo
- **Clay Mathematics Institute**, full funding for the workshop *Equivariant methods in geometry*, University of Ibadan, 2024
- **MATH+ travel grants** (Makerere, AIMS Ghana) and **ICTP** funding (Trieste 2019 and 2022, Dar-es-Salaam 2018)

- **Invited participant**, Isaac Newton Institute programmes *New Equivariant Methods in Algebraic and Differential Geometry* (Cambridge, 2024) and *Algebraic Groups, Geometry, Invariants and Related Topics* (Essex, 2026)

Selected talks

- *Finite automorphisms of toric orbifold hypersurfaces*. AGGITatE 2026, Isaac Newton Institute satellite programme, Essex, 2026
- *The Frobenius problem, simplices and moduli spaces*. Mathematics and Statistics Colloquium, The Open University, 2025
- *Families of hypersurfaces from polytopes*. *Algebra and Geometry from Africa*, ICMS Edinburgh, 2025
- *Geometry of some moduli spaces of hypersurfaces*. COW Seminar, University of Bath, 2024
- *Moduli of coverings branched along a hypersurface*. Isaac Newton Institute, Cambridge, 2024
- *Stacky fans and tropical moduli in polymake*. ISSAC 2023, Tromsø
- *Projective moduli of semistable hypersurfaces*. Geometry and Topology Seminar, Universität Wien, 2023

A full list of 19 talks is in the long CV.

Teaching and supervision

- Lecture courses at TU Berlin (Algebra I & II, Algebraic Geometry, Analysis for Engineers), Makerere University, AIMS Ghana and the Open University
- Invited lecture courses: lattice polytopes (ICMS Edinburgh, 2025); moduli spaces and group actions (University of Ibadan, 2024); homological algebra (EAALG, 2021)
- Supervision at TU Berlin: three completed Bachelor's theses; two Master's and two Bachelor's theses in progress

Service and organisation

- Referee for *Bull. LMS*, *J. LMS*, *IMRN*, *Advances in Geometry*
- Programme committee, Young African Mathematicians fellowship, MATH+, Berlin; international collaborator, Eastern Africa Algebra Research Group
- Outreach: *Calculate with Africa*, Meridian podcast, 2024; public talks *Why mathematics?* in Kampala, 2022
- Organiser: workshops at Makerere University (2021, 2027 forthcoming), University of Ibadan (2024), TU Berlin (2020); group leader, cubic surfaces workshop, University of Oslo (2019)

Languages

English (native) · German (fluent) · Luganda (beginner)